

BHANU SHARMA

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EDUCATION

MS in Bioinformatics (Data Science Focus) (GPA: 4.0)

Northeastern University

Sept 2024 - Dec 2026

Boston, MA, USA

Relevant coursework: Deep Learning, Foundations of AI, ML, Statistics, Algorithms

Bachelor of Technology in ECE (GPA: 7.7/10)

Punjab Engineering College (PEC)

Aug 2016 - May 2020

Chandigarh, India

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, UNIX/Shell, Scala, Java, SAS

Libraries: Pytorch, Tensorflow, Numpy, Pyspark, Pandas, Matplotlib, ggplot2, scikit-learn

AI Tools: vLLM, TRT-LLM, AIPerf, Openclaw, Zeus Energy

Tools: Jupyter, Power BI, Tableau, Azure, Databricks, Snowflake, Spark, Docker, ADF, AWS, MS Office, Github

Skills: Machine Learning, AI, Natural Language Processing (NLP), Benchmarking, Data Visualization, Big Data, Data Warehousing, Business Intelligence, ETL pipelines, CI/CD, Agile, Cloud

EXPERIENCE

Dell Technologies, Hopkinton, MA: AI Research Intern

Jan 2026 - June 2026

- Developed LLM inference simulator that, given hardware/model specs and SLA targets, evaluates serving configs to predict throughput, latency, TTFT, and ITL within $\pm 15\%$ **accuracy at batch sizes of 512+** giving optimal config for vLLM serving setup, incorporating continuous batching, parallelism, and disaggregated serving.
- Extended the simulator to support **Mixture-of-Experts (MoE)** models and multiple quantization strategies, enabling efficient and cost-aware inference optimization saving a projected **1500+ GPU hours**
- Modeled and Implemented Novel LLM Inference Energy and Power estimator based on roofline model of energy, achieving up to **90% accuracy**

Northeastern University, Boston, MA: Teaching Assistant

Sept 2025 - Present

- Assisted Professor Valerie Hower with MATH 7340: Statistics for Bioinformatics by conducting weekly office hours and supporting **90+ graduate students** in R programming and core statistical concepts

UnitedHealth Group (Optum), Gurgaon, India: Data Engineer

Aug 2020 - Jun 2024

- Migrated legacy ETL pipeline to Spark/Scala and Snowflake on Azure, reducing database **costs by up to 90%**
- Redesigned ETL architecture using PySpark, MongoDB, Azure Databricks, and ADF, handling **3TB+ of daily data**
- Built Power BI dashboards delivering insights to **50+ clients across 4 departments**
- Developed Python-based alerting system for automated KPI reporting, saving **6+ hours/week**

PUBLICATIONS

Operational Resilience Under Carbon Constraints: A Multi-Agent Approach: [DOI](#)

March 2026

- Developed a multi-agent system leveraging LLM orchestration and predictive analytics to model carbon–cost–service trade-offs in global supply chains

Some Patterns of Sleep Quality and Daylight Savings Time Across Countries: [DOI](#)

Aug 2025

- Analyzed sleep quality and DST patterns across 61 countries using predictive modeling and exploratory ML

PROJECTS

Swaad - Flavor Profile Based Food Recommendation System - [Github](#)

Dec 2025

- Built an AI-powered food recommendation system that uses Gemini Flash multimodal LLM to extract dishes from menu images, generates per-dish embedding vectors, and applies KNN similarity matching against a user taste profile built across 1,000+ ingredients to deliver real-time personalized recommendations.

Rating Aware Beer Recommendation System - [Github](#)

Aug 2025

- Built a recommendation system using KNN and Gradient Boosted Trees with LLaMA-based NLP interface to interpret natural language queries and recommend beers based on flavor profiles and predicted ratings